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Smythe, Nathan C.
Hanson, Susan Meriel Kloek
Hudston, Lisa Ann

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Title Page

Nathan C. Smythe[1], Susan K. Hanson[1], Lisa A. Hudston[1]

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[1]Nuclear and Radiochemistry, Los Alamos National Laboratory, 87545, Los Alamos, USA

nsmythe@lanl.gov

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Nathan C. Smythe*[1], Susan K. Hanson[1], Lisa A. Hudston[1]

[1]Nuclear and Radiochemistry, Los Alamos National Laboratory, 87545, Los Alamos, USA

Abstract

The long-lived fission product ^{151}Sm ($T_{1/2} \approx 90$ y) present in archived dissolved Trinity nuclear test debris samples was analyzed using liquid scintillation counting and the results were compared to the expected values based upon historical analysis results. The measured ^{151}Sm activities were found to agree with the expected values within experimental error. This demonstration validates the use of this technique in situations where the sample is too old for traditional shorter half-life radioactive fission product signatures such as ^{99}Mo ($T_{1/2}$ = 2.75 d), ^{97}Zr ($T_{1/2}$ = 16.7 h), ^{147}Nd ($T_{1/2}$ = 10.98 d), and ^{153}Sm ($T_{1/2}$ = 46.3 h), or too dilute for determining fissions through stable isotope ratio measurements.

Keywords

Liquid scintillation, ^{151}Sm, nuclear forensics, Trinity

Introduction

Forensic examination of near-term nuclear events takes advantage of many short-lived radio-isotopes to facilitate analysis and assessment. As time between the event and the analysis increases, beta and gamma counting of these isotopes becomes infeasible due to decay. Consequently, other approaches and/or analytes are necessary in order to analyze older samples. For example, the examination of stable isotope ratio perturbations as has been done in a recent reassessment of the Trinity nuclear test.[1] While successful and useful, this technique requires a threshold of initial fissions in order for the measurement to be made. Thus, there is room in the forensic toolbox for complimentary techniques that are sensitive to both older and low-level samples.

One approach for these types of samples is the analysis of ^{151}Sm (β-100 % $T_{1/2} \approx 90$ years, β_{max} = 76.3 keV) due to its longer half-life as compared to ^{153}Sm ($T_{1/2}$ = 46.3 h) and the ability to make use of already well-developed chemical separation procedures. As ^{151}Sm does not occur naturally, its detection is affirmation of a man-made source and weak initial activity can be compensated for by increasing the amount of material analyzed. The challenges in performing such an analysis are that the longer half-life results in relatively fewer disintegrations per minute (dpm) and the relatively weaker beta energy (approximately a factor of 10 less than ^{153}Sm) further lowers the signature associated with ^{151}Sm decay. These properties suggest an approach based on liquid scintillation counting (LSC) as samples of this type are well within the capabilities of the

technique. This approach also allows for the possibility to use natural ^{147}Sm (α-100%, $T_{1/2}$ = 1.06 x 10^{11} y) present in the carrier for in-situ yield determination.

In this paper, we apply the lessons of previous work on ^{151}Sm LSC [2] to the analysis of enriched ^{235}U (HEU) exposed to thermal neutrons for method validation, and then to Trinity nuclear test debris samples. The results are compared with those expected based on canonical ^{99}Mo and ^{97}Zr fissions respectively. We also evaluate both gravimetric and ^{147}Sm LSC yield determinations and examine the sensitivity limit of the method.

Experimental

The ^{235}U fission solution samples consisted of an HEU foil that had been irradiated at the MIT reactor for 200 minutes at 90 kW power. The irradiated HEU foil was dissolved in 4 M HNO_3 upon gradual addition of concentrated HCl while heating gently on a hotplate set at 100 ^{o}C. The final HEU foil solution was prepared in 4 M HNO_3 and analyzed using standard radiochemistry procedures for thermal calibration (TCal) experiments.[3] All debris samples were obtained from LANL internal archives as stock solutions with known mass (g debris per g solution) ratios and had been previously analyzed.[1,4] The Trinity nuclear debris samples were dissolved by repeated cycles of heating to dryness in a 1:1 mixture of concentrated HF and HNO_3 and stored as final solutions in 3 M HCl (with a final concentration of ~8 mg solid per gram solution). One solution (Trinity-4) was analyzed in triplicate using independent aliquots (Trinity-4.1, Trinity-4.2, Trinity-4.3) in order to test reproducibility. For every analysis, a weighed aliquot was taken and a calibrated aliquot of certified natural isotopics Sm solution (SPEX CertiPrep, 5% HNO_3 or HCl solution and nominally 10.0 mg mL^{-1}) targeting ca. 1 mL was added. Every sample was subjected to a standard two-step purification process consisting of: (1) chemical operations designed to yield a purified lanthanide mixture (minus Ce, which is removed as Ce(IV)) and (2) HPLC chromatography designed to separate individual lanthanide elements.[3]

The HPLC system consists of Agilent 1260 Infinity II quaternary pumps attached to Rheodyne 3725i sample injectors utilizing 10 mL sample loops and Omnifit 6mm columns packed with ca. 25.4 cm of Dowex 50W-X8 resin. The elution solvent was a 1 M *α*-hydroxyisobutyric acid (HIBA) solution adjusted to ca. pH 5.3 using NH_4OH and used in a gradient from 8-12 % at a flow rate of 3.5 mL min^{-1}. Fractions were collected at 1 minute intervals and the Sm-containing tubes were identified via precipitation with oxalic acid. The fractions containing Sm were combined (culling the wings), additional oxalic acid added to ensure complete precipitation, and a small amount of shredded Whatman filter paper prior to gentle heating to ensure coagulation. The solids were subsequently collected and burned in a ceramic crucible at 1100 °C for at least 30 minutes to generate Sm_2O_3.

The resulting Sm_2O_3 was collected on a pre-weighed filter paper and washed with EtOH. After drying, the samples were weighed on a Mettler Toledo XS105 DualRange balance (41 g / 120 g range with d = 0.01 mg / 0.1 mg respectively) to determine the gravimetric yield. Once yielded, the samples were again burned and the Sm_2O_3 was dissolved in concentrated HCl (Fisher, trace metal grade) which was evaporated using gentle heating. The resulting residue was then re-

dissolved in 2 mL of 2 M HCl and transferred to a vial suitable for liquid scintillation counting. Immediately before counting, 13 mL of LSC cocktail (Ultima Gold AB) was added and the vial agitated to ensure mixing.

Radiometric analysis was performed on a PerkinElmer Quantulus 1220 ultra low-level Liquid Scintillation Spectrometer counter. The instrument utilizes dual low noise photomultiplier tubes as well as asymmetric passive lead shield and an asymmetric guard counter which operates in anticoincidence with the sample detector to significantly reduce background signal. To determine each sample's quench parameter, an external ^{152}Eu source is exposed to the sample for 1 minute at the beginning of the count, then retracted while the count is paused for 1 minute. After that each sample is counted as detailed below.

For counting, each run also contained vials consisting of: (1) 2 mL of 2 M HCl as a background, (2) A calibrated quantity of Sm solution as a ^{147}Sm standard, and (3) A calibrated quantity of ^{151}Sm standard (Eckert & Ziegler). The Pulse Shape Analysis (PSA) setting was found to give good sensitivity for alpha at 50 and beta at 141 when tested against the Sm solution and ^{151}Sm standard respectively, so samples and background were counted using both settings and the respective standards were used as appropriate. A 500 minute count time was used for all counts except where noted.

Results

The concentration of ^{151}Sm was initially measured in two different samples of irradiated HEU metal. These thermal calibration (TCal) samples represent a well-defined material exposed to a well-defined irradiation environment and make an excellent benchmark (the ^{99}Mo fissions values per g of initial solution can be measured to a high level of fidelity). Following that, the ^{151}Sm concentration was also measured in five different Trinity nuclear debris samples, one of which, Trinity-4, was measured in triplicate in order to test the reproducibility of the LSC measurement across independent analyses. Table 1 summarizes the ^{151}Sm measurements and their comparison to expected values based on the total fissions as determined by measurement of the fission products ^{99}Mo (TCal) or ^{97}Zr (Trinity). This comparison is made using exponential decay function (1) and the appropriate ^{151}Sm cumulative fission chain yield.[5] Multiplication of the appropriate fissions g^{-1} solution value with the appropriate ^{151}Sm cumulative fission chain yield yields an expected atoms g^{-1} solution value that can be compared with the experimentally determined value.

$$A_t = A_0 e^{\left(\frac{-ln(2)t}{T_{1/2}}\right)} \tag{1}$$

A_0 = Reference Activity, t = Time Interval, $T_{1/2}$ = Half-Life

These results are based on gravimetric yielding of Sm_2O_3 and the longer established ^{151}Sm half-life of 90(8) years [6,7] as a baseline. All of the results afford good agreement with the expected values, and the three independent Trinity-4 analyses agree with each other within error. Overall, this shows that even in this baseline approach, the method is both reproducible and accurate. In the discussion sections below, we examine the potential limits of the LSC technique. The impact

of yielding via ^{147}Sm [2] and an alternate ^{151}Sm half-life of 94.7(6) years [8,9] on the results are also probed. A full table of the results from these approaches can be found in the supplementary information.

Table 1 Summary of measurements of thermal calibration (TCal) and Trinity debris (Trinity) samples derived from a ^{151}Sm half-life of 90(8) [6,7] and gravimetric yielding. TCal results are compared to ^{99}Mo fissions g^{-1} and Trinity results are compared to ^{97}Zr fissions g^{-1} using ^{235}U thermal and ^{239}Pu fission chain yields respectively.[5] The reference dates of the TCal and Trinity samples are October 19, 2018 and July 16, 1945 respectively.

	Measured (dpm)	**Time Int. (d)**	**Ref. Date Conc. (atoms/g)**	**Expected Conc. (atoms/g)**	**Expected / Measured**
TCal-1	72(3)	757	4.4(4)E9	4.54(10)E9	1.04(10)
TCal-2	75(3)	758	4.3(4)E9	4.54(10)E9	1.06(10)
Trinity-1	265(8)	27722	2.6(3)E10	2.65(5)E10	1.01(11)
Trinity-2	305(10)	27722	3.0(3)E10	3.08(6)E10	1.04(11)
Trinity-3	188(6)	27723	1.95(21)E10	1.95(4)E10	1.00(11)
Trinity-4.1	216(8)	27516	2.5(3)E10	2.56(5)E10	1.04(11)
Trinity-4.2	179(7)	27517	2.5(3)E10	2.56(5)E10	1.03(11)
Trinity-4.3	211(7)	27715	2.4(3)E10	2.56(5)E10	1.08(12)
Trinity-5	153(5)	27724	1.55(17)E10	1.56(3)E10	1.01(11)

Discussion

Half-life and Half-life Uncertainty

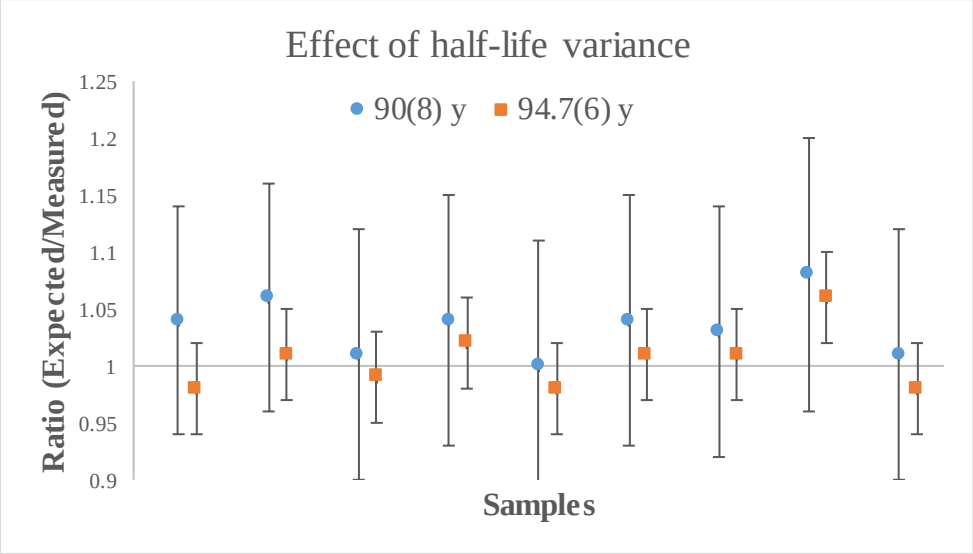

Fig. 1 Expected/measured ratios of the TCal and Trinity results along with 1-σ errors derived from ^{151}Sm half-life values of 90(8) and 94.7(6) years.

Two different values of the ^{151}Sm half-life have been adopted by ENDF/BVIII.0 (90(8) y) [6] and JEFF-3.3 (94.7(6) y) [9], the latter of which derives in particular from a more recent measurement.[8] As shown in Fig. 1, the magnitude difference between the results obtained from these two half-life values does not materially affect the results for the time intervals of the TCal (ca. 2 years) and Trinity (ca. 76 years) samples. This is consistent with the calculation that it would require a time interval of ca. 276 years to achieve a 0.9 ratio in the decay correction. So, in the context of nuclear forensics as well as any other relatively short-termed analysis, the magnitude difference is not especially significant. What does have significant implications is the nearly order of magnitude difference in uncertainty. Full error propagation for equation (1) is given in equation (2). Assuming that the time interval uncertainty can be ignored, (2) can be simplified to equation (3) to give the fractional uncertainty of the calculated activity at the reference time. As would be expected, the half-life uncertainty contribution increases as the time interval increases. However, the conversion of dpm g^{-1} to atoms g^{-1} also depends upon the half-life (4). Combining these, (5) represents the total propagation from measurement to comparison.

$$\sigma_f = \left[\sigma_{A_0}^2 e_0^2 + \sigma_{t_{1/2}}^2 \left(A_0 e_0 \frac{ln(2)t}{T_{1/2}^2}\right)^2 + \sigma_t^2 \left(A_0 e_0 \frac{-ln(2)}{T_{1/2}}\right)^2\right]^{1/2}, \; e_0 = e^{\left(\frac{-ln(2)t}{T_{1/2}}\right)} \tag{2}$$

$$\frac{\sigma_f}{A_0 e_0} \approx \left[\frac{\sigma_{A_0}^2}{A_0^2} + \frac{\sigma_{t_{1/2}}^2}{T_{1/2}^2}\left(\frac{ln(2)t}{T_{1/2}}\right)^2\right]^{1/2} \tag{3}$$

$$A_{atoms} = \frac{A_{dpm} \times T_{1/2}}{ln(2)} \tag{4}$$

$$\frac{\sigma_f}{A_0 e_0} \approx \left[\frac{\sigma_{A_0}^2}{A_0^2} + \frac{\sigma_{t_{1/2}}^2}{T_{1/2}^2}\left[\left(\frac{ln(2)t}{T_{1/2}}\right)^2 + 1\right]\right]^{1/2} \tag{5}$$

For a hypothetical measurement with a 5% uncertainty in the ^{151}Sm standard, the propagated errors from the two half-life values as a function of time interval are shown in Fig 2. The largest contributing factor to the total uncertainty when using a ^{151}Sm half-life of 90(8) is the half-life uncertainty while the 5% standard uncertainty dominates when using a half-life of 94.7(6). This clearly shows that how well the half-life is known can have a profound effect on the conclusions that can be drawn (e.g. see discussion of Fig. 4 below), and that one of the keys to more precise measurements is knowing this value as precisely as possible. While the values in Fig. 1 would seem to suggest that 94.7 years is closer to the true value of the ^{151}Sm half-life, caution should be taken before reaching such a conclusion as these results can also be impacted by error in the fission chain yields.[10,11]

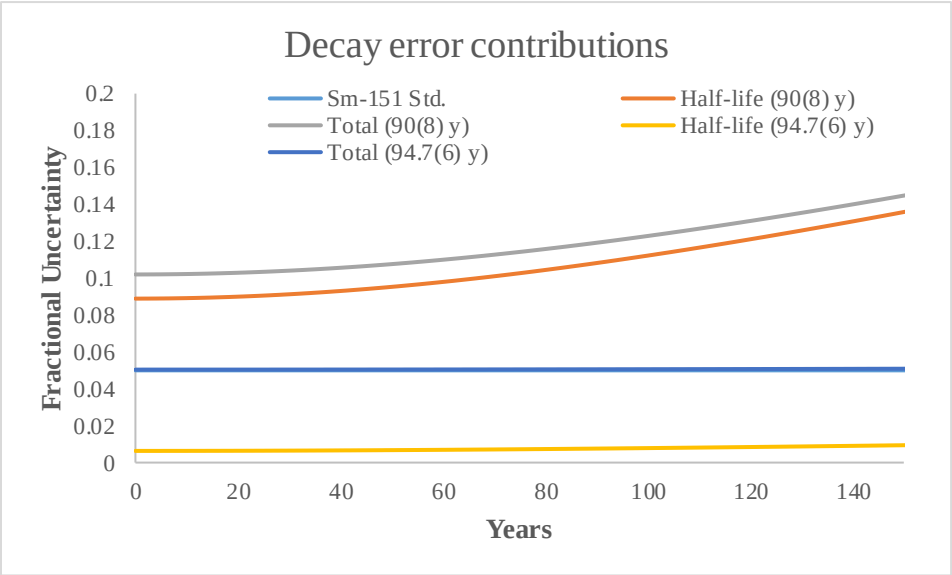

Fig. 2 Evolution of the total uncertainty of a hypothetical ^{151}Sm analysis vs. the number of years between the event and analysis assuming a 5% uncertainty in the reference activity.

Gravimetric vs. LSC Yields

To accurately calculate the total amount of ^{151}Sm in a sample, the observed activity in the LSC measurement must be corrected for the chemical yield of the purification. Based on extensive experience in conducting radiochemical measurements in our laboratories, calculating yields based upon gravimetric measurements is highly effective and accurate. However, weighing the oxide requires extra manipulative steps not needed for solution counting. As detailed previously, an alternate method based on LSC of ^{147}Sm would obviate the need for isolation of Sm_2O_3 and shorten the measurement time.[2] While the LSC yields determined in the previous work were generally consistent with the gravimetric values, they were offset from each other. With a number of additional measurements made in the course of this study, we examine potential causes of this difference.

Fig. 3 shows that there is a clear trend of the LSC-determined yields being higher than gravimetric yields. The lack of apparent correlation of this difference to the magnitude of the yield or to the measured activity suggests that this difference does not derive from an inherent bias between the techniques, but rather has at least some contribution from the sample and/or sample preparation. This is further supported by control measurements that showed a very close to linear response to ^{147}Sm concentration variations (supp. Fig. 1) and that ^{151}Sm beta cross-talk in the ^{147}Sm alpha data is minimal and non-overlapping (supp. Fig. 2, [2]).

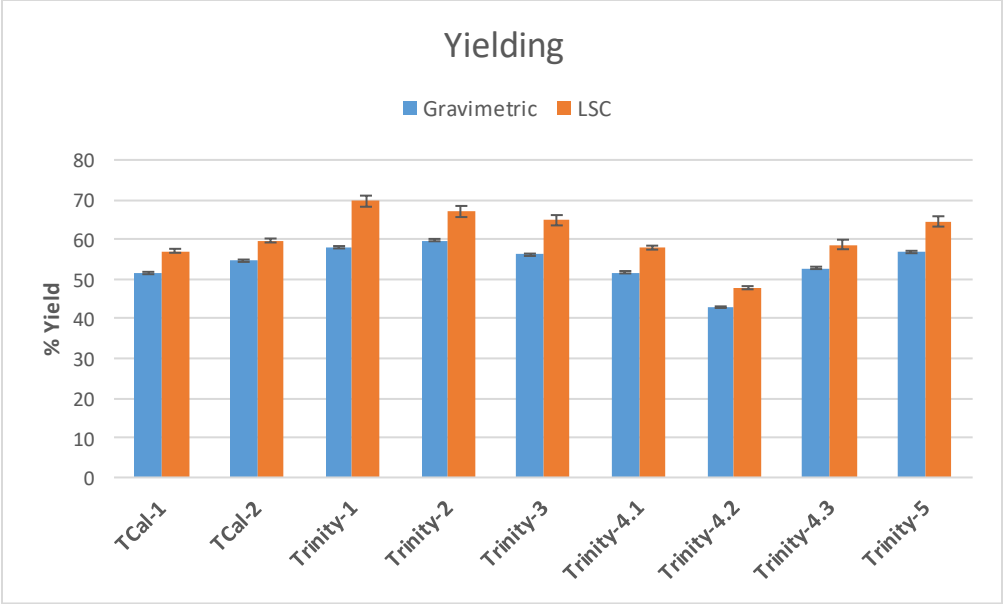

Fig. 3 Comparison of chemical yields determined gravimetrically vs. ^{147}Sm LSC.

If the differences were caused by Sm contamination in the reagents and materials used, it would require acquiring an additional 0.5 mg at the end of processing to achieve a 10% offset assuming a 50% yield. This quantity would be detectable in gravimetric yielding, and nothing above background was found in the stock reagents and materials by LSC. It also cannot be coming from the Trinity debris because the effect was also seen in the TCal HEU samples. Isotopic contamination (see supporting information for more details) would seem to be a likely possibility given the long half-life of ^{147}Sm, but there does not appear to be a likely candidate that has a relatively short half-life, is a close enough neighbor of Sm to be potentially present after purification, and is accessible in the fission chain.

Quantitative Effect of Yield Difference

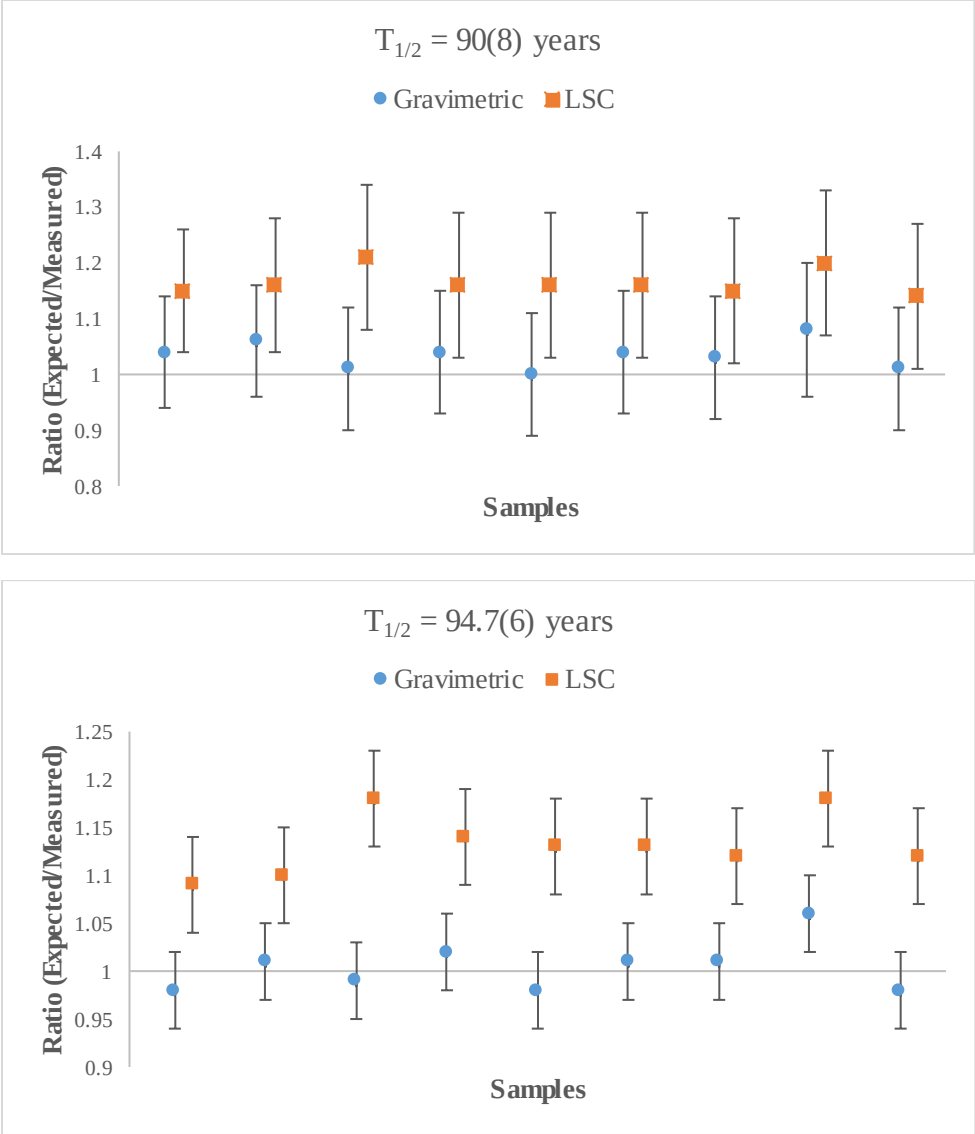

Fig. 4 Summary of ratios of expected to measured activity for gravimetric and LSC yielding based on a ^{151}Sm half-life of 90(8) y (top) and 94.7(6) y (bottom) and the 1-σ uncertainty. LSC yield results are offset to the right compared to the corresponding gravimetric yield results.

The difference in expected/measured ratios between results from gravimetric and LSC yielding may or may not be considered significant depending upon the uncertainty of the half-life. With the larger 8 year uncertainty (Fig. 4, top), they could be considered similar within a 1-σ margin, but with the LSC magnitude higher than expected due to higher determined yield. This changes with the 0.6 year uncertainty (Fig. 4, bottom), where 2-3-σ would have to be used to consider the approaches equivalent. Based on this, our long experience in conducting consistent analyses using gravimetric yielding, and the consistent results with expected values when using gravimetric yielding, we consider this to be the most accurate method at this time. That is not to say that yield determination via LSC gives wholly wrong or inconsistent results, merely that it is slightly less

accurate within this context. This is unexpected given that an in-situ measurement should provide less opportunity for manipulative losses (i.e. collecting, weighing, and redissolving the sample). But if isotopic contamination is occurring, it would point to the need to refine our purification methodology for these more low-level measurements rather than to a deficiency in the LSC yielding technique itself.

Method Sensitivity

For the samples examined herein, aliquot sizes were chosen to ensure that there would enough activity for a facile analysis. But, an important component of a nuclear forensic capability is the ability to analyze samples that may only exhibit weak activity. To determine the detection limit, we examined the measured vs. expected activity of decreasingly small amounts of the ^{151}Sm standard (Fig. 5). The measurements (shown in orange) are all within at most 0.3 dpm of the expected values. This excellent agreement was measured down to 5.0 dpm and the deviation of each measurement is less than the uncertainty of the standard itself. This indicates that reliable measurements should be possible down to at least 7 dpm accounting for an ca. 77(2) % effective efficiency (supp. Table 2). Given a set of background measurements showing a background of 4.3(2) counts min^{-1}, a 95 % confidence level suggests a possible minimum detectable quantity of ca. 0.5 dpm if stringent care is taken in preparation, handling, and counting of samples.

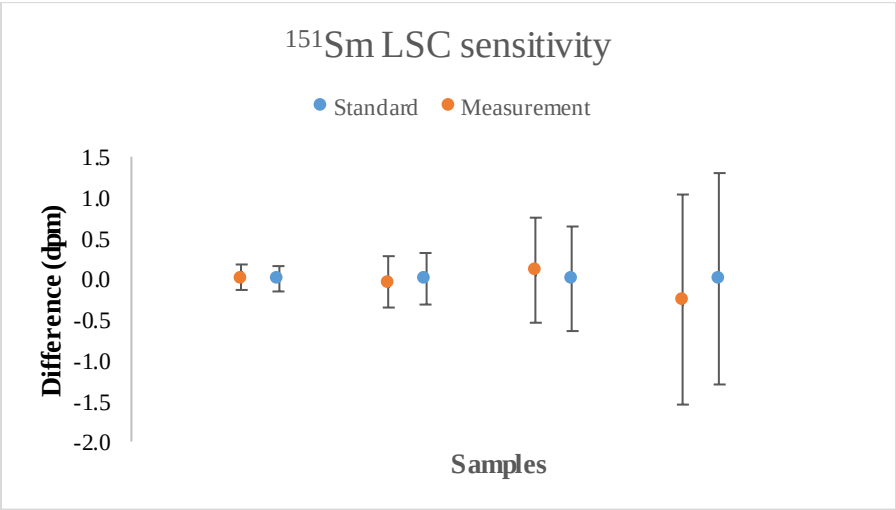

Fig. 5 Difference between measured (orange dots) and expected (blue dots) activity of varying quantities of ^{151}Sm standard. Left to right, expected activities are 5.0, 10.2, 20.7, and 41.9 dpm with uncertainties of 3.1 %.

Conclusions

Both HEU and Trinity debris samples have been examined using ^{151}Sm liquid scintillation counting. The results of these measurements show excellent agreement with the expected results based upon other, well-established, analyses. For the timeframes examined, this agreement stands

regardless of the absolute value of the ^{151}Sm half-life chosen. It was also established that accurate analyses of samples down to at least 7 dpm can be expected with a lower threshold possible. Based on previous analytical experience, we have a preference for gravimetric yielding although ^{147}Sm internal tracer yielding should be equally valid. The offset in yields obtained when comparing both methods currently lack an obvious explanation and merits further investigation. This study serves as a demonstration of the utility of LSC in examining samples of medium age in terms of nuclear forensics.

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Declarations

Conflict of Interest: The authors have no financial or non-financial conflicts of interest to declare.

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Supplementary Information

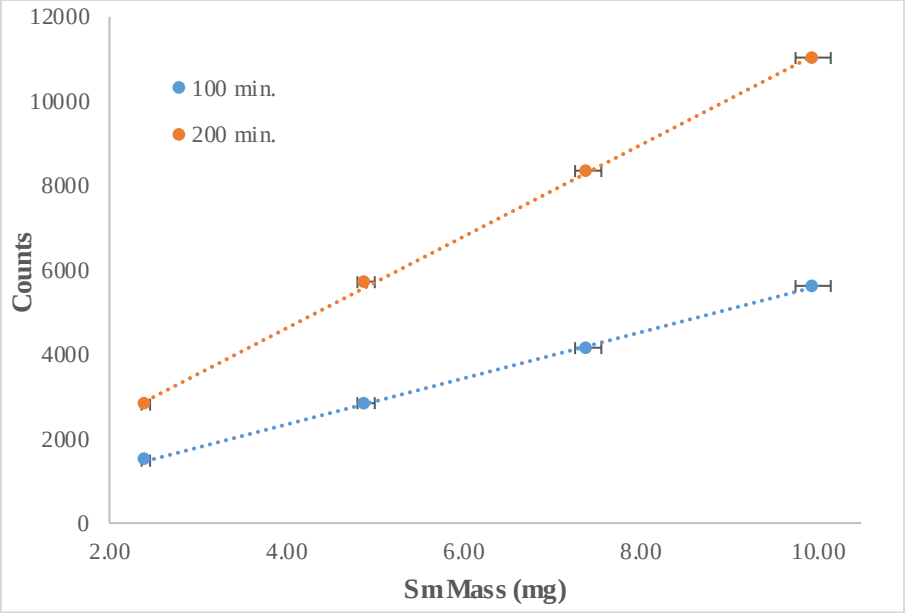


Fig. 1 LSC measurement of ^{147}Sm from the carrier solution in decreasing quantity with counting at 100 and 200 minutes.

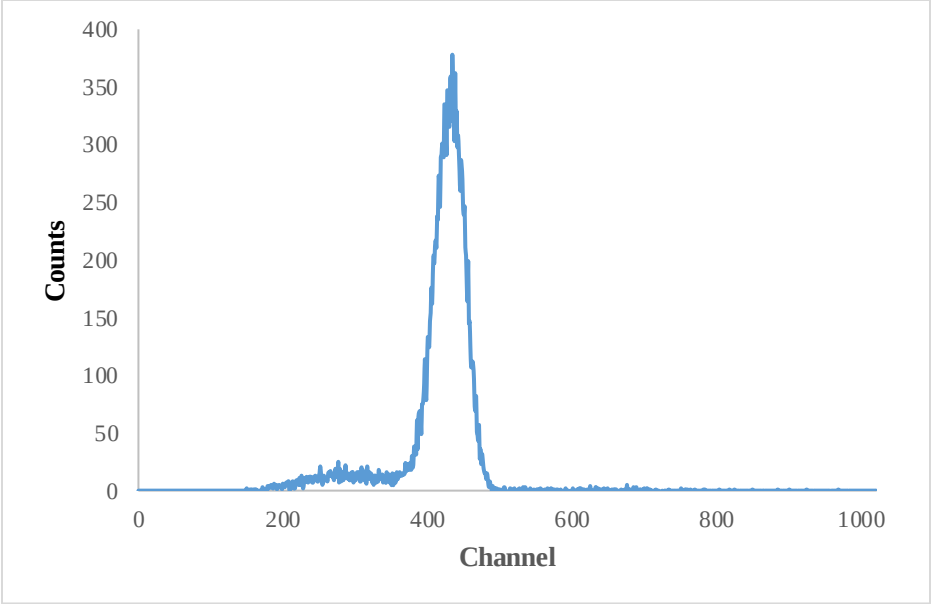
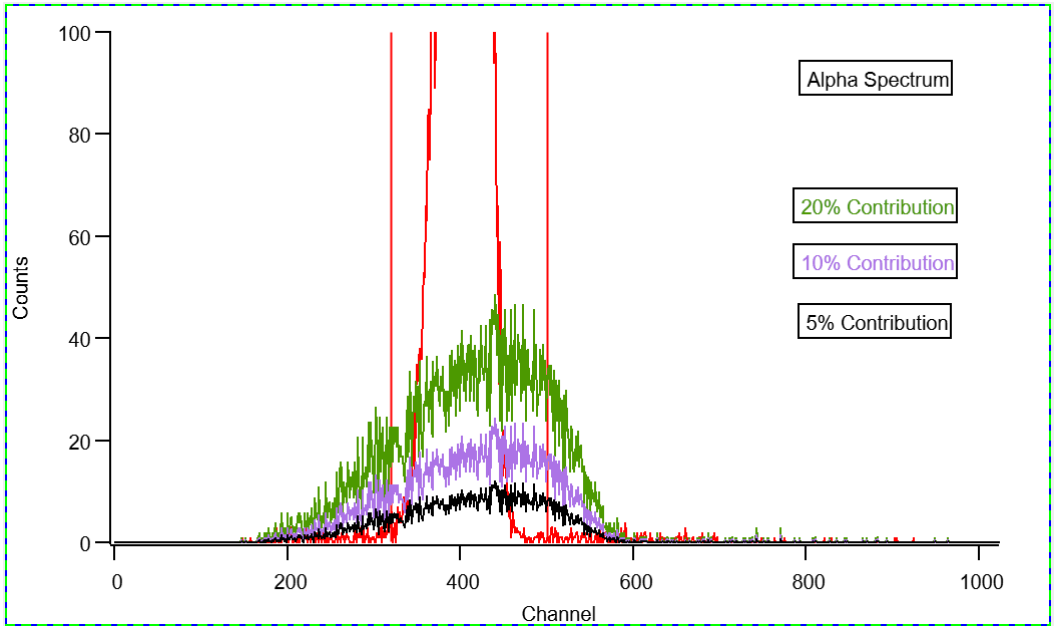


Fig. 2 Alpha spectrum of Trinity-3 at PSA 50 showing primarily ^{147}Sm (right) with a small amount of ^{151}Sm (left) from beta cross-talk.



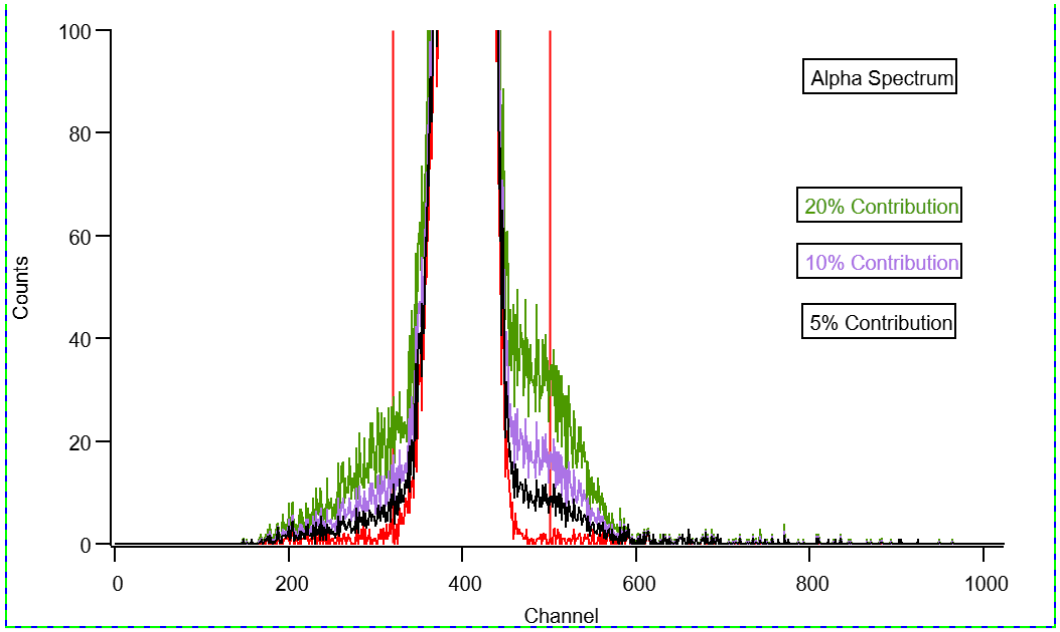


Fig. 3 Overlay of ^{147}Pm beta cross-talk over ^{147}Sm at 5, 10, and 20 % of the ^{147}Sm signal (top) and modeling of the addition of the ^{147}Pm to the ^{147}Sm (bottom). Any significant ^{147}Pm cross-talk contribution would be evident in the peak shape. Vertical red lines are graph markers.

Comments on Isotopic Contamination

Contamination of LSC yielding measurements by ^{147}Sm rather than Sm is not a likely source of the higher LSC yield determinations. Given the 14.99% abundance of ^{147}Sm in natural Sm, it would require 0.07 mg of ^{147}Sm to have the same effect as 0.5 mg of Sm. While this quantity would be less visible in the gravimetric yield determinations (ca. 1 % vs. 10 % mass change), it would require a significant source of pure ^{147}Sm. Consider the TCal samples which originally had a ^{147}Nd content of ca. 2.5 x 10^{10} atoms/g solution. Given an aliquot of ca. 2.2 g, that is an initial ^{147}Nd loading of 5.5 x 10^{10} atoms. Even if all of the ^{147}Nd had decayed to ^{147}Sm (a significant overestimation), this represents merely 1.3 x 10^{-8} mg; too small of a potential contribution by a factor ca. 10^{6}. In terms of other Sm isotopes, the ^{148}Sm half-life is too long to be of consequence, and ^{146}Sm should not be accessible in the fission chain. Other contaminating isotopes are possible such as those of nearest-neighbors Pm and Eu as well as Am. ^{147}Pm cross-talk can be excluded because peak shape analysis shows that it is not present in the TCal samples (supp. Fig. 3) and the Trinity debris samples are too old to harbor any significant quantities of this isotope ($T_{1/2}$ = 958.19 d). Eu does not appear to have a good candidate for interference, and the alpha radiation energies of ^{243}Am and ^{241}Am occur at more than twice the energy than that of ^{147}Sm. Ultimately, we cannot propose a likely candidate contaminant that would have enough activity to affect the ^{147}Sm alpha measurement and enough elution overlap with Sm at this time. But, this mode of interference has been observed for other separations such as ^{91}Y/^{161}Tb, where very small amounts of higher-activity ^{91}Y co-eluting with ^{161}Tb can interfere with beta measurements of the much weaker ^{161}Tb activity.

Table 1 Summary of measurements of thermal calibration (TCal) and Trinity debris (Trinity) samples using ^{151}Sm half-life values of both 90(8) (ENDF/B-VIII.0) and 94.7(6) (JEFF-3.3) years. Only unique values are shown. Thermal calibration results are compared to ^{99}Mo fissions g^{-1} and Trinity results are compared to ^{97}Zr fissions g^{-1} using ^{235}U thermal and ^{239}Pu fission chain yields respectively. Grav-* and LSC-* refer to gravimetric yielding and LSC yielding respectively. 90 and 95 refer to the ENDF and JEFF ^{151}Sm half-life values. The reference dates of the TCal and Trinity samples are October 19, 2018 and July 16, 1945 respectively.

	Yield (%)	Measured (dpm)	Time Int. (d)	Ref. Date Conc. (atoms/g)	Expected Conc. (atoms/g)	Expected / Measured
TCal-1						
Grav-90	51.5(3)	72(3)	757	4.4(4)E9	4.54(10)E9	1.04(10)
LSC-90	57.1(5)			3.9(4)E9		1.15(11)
Grav-95	51.5(3)	73(3)		4.62(17)E9		0.98(4)
LSC-95	57.1(5)			4.17(16)E9		1.09(5)
TCal-2						
Grav-90	54.6(3)	75(3)	758	4.3(4)E9	4.54(10)E9	1.06(10)
LSC-90	59.7(5)			3.9(4)E9		1.16(12)
Grav-95	54.6(3)			4.52(17)E9		1.01(4)
LSC-95	59.7(5)			4.13(16)E9		1.10(5)
Trinity-1						
Grav-90	58.0(3)	265(8)	27722	2.6(3)E10	2.65(5)E10	1.01(11)
LSC-90	69.6(14)			2.19(24)E10		1.21(13)
Grav-95	58.0(3)			2.69(9)E10		0.99(4)
LSC-95	69.6(14)			2.24(9)E10		1.18(5)
Trinity-2						
Grav-90	59.8(3)	305(10)	27722	3.0(3)E10	3.08(6)E10	1.04(11)
LSC-90	67.0(14)			2.7(3)E10		1.16(13)
Grav-95	59.8(3)			3.03(10)E10		1.02(4)
LSC-95	67.0(14)			2.71(10)E10		1.14(5)
Trinity-3						
Grav-90	56.1(3)	188(6)	27723	1.95(21)E10	1.95(4)E10	1.00(11)
LSC-90	64.8(13)			1.69(18)E10		1.16(13)
Grav-95	56.1(3)			1.99(6)E10		0.98(4)
LSC-95	64.8(13)			1.72(7)E10		1.13(5)
Trinity-4.1						
Grav-90	51.7(3)	216(8)	27516	2.5(3)E10	2.56(5)E10	1.04(11)
LSC-90	57.9(5)			2.21(24)E10		1.16(13)
Grav-95	51.7(3)	217(8)		2.54(10)E10		1.01(4)
LSC-95	57.9(5)			2.27(9)E10		1.13(5)
Trinity-4.2						
Grav-90	42.9(2)	179(7)	27517	2.5(3)E10	2.56(5)E10	1.03(11)
LSC-90	47.8(4)			2.23(24)E10		1.15(13)
Grav-95	42.9(2)	180(7)		2.55(10)E10		1.01(4)
LSC-95	47.8(4)			2.29(9)E10		1.12(5)

Trinity-4.3						
Grav-90	52.8(3)	211(7)	27715	2.4(3)E10	2.56(5)E10	1.08(12)
LSC-90	58.7(12)			2.13(23)E10		1.20(13)
Grav-95	52.8(3)			2.42(8)E10		1.06(4)
LSC-95	58.7(12)			2.18(8)E10		1.18(5)
Trinity-5						
Grav-90	56.8(3)	153(5)	27724	1.55(17)E10	1.56(3)E10	1.01(11)
LSC-90	64.5(13)			1.37(15)E10		1.14(13)
Grav-95	56.8(3)			1.59(5)E10		0.98(4)
LSC-95	64.5(13)			1.40(5)E10		1.12(5)

Table 2 Counting stability of an aliquot of ^{151}Sm standard with an average counting efficiency of 77 ± 2%

	Counts	**Counting Time (min)**	**Counts/min**
Run 1	32730	250.03	130.91
Run 2	33101	250.03	132.39
Run 3	32627	250.03	130.49
Run 4	32962	250.03	131.83
Run 5	32645	250.03	130.57
mean	32813		131.2
stddev	209		0.8